Unit 5 CARBOHYDRATES

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Gerty Cori

Gerty Cori along with her husband Carl Cori made important discoveries in Carbohydrate biochemistry. They were particularly interested in glucose metabolism and its hormonal regulation. Their proposed cycle of reactions is now known as Cori's cycle. They were awarded Nobel prize in 1947 for their discoveries. Gerty Cori was the first American woman to receive nobel prize.

The US government has also released an US Postal Service stamp in Honour of Gerty Cori. The Cori craters in the moon and venus were also named after them.

Learning Objectives

After reading this Chapter, you will be able to:

- Explain the functions of carbohydrates
- Draw the chemical structure of glucose, galactose, fructose, maltose, lactose and sucrose.
- Write the physical and chemical properties of carbohydrates
- Draw the Haworth projection of glucose, galactose and fructose
- Describe the structural arrangement of the Homo polysaccharides (Starch & Glycogen) and Hetero polysaccharides (Heparin and hyaluronic acid)

Introduction

Carbohydrates are widely distributed in plants and animals. Plants produce carbohydrates from carbon dioxide and water by photosynthesis. The important carbohydrate found in plants is starch. Animals mainly depend upon plant sources for their carbohydrates. The carbohydrate stored in animals is glycogen.

Chemically carbohydrates are defined as polyhydric aldehyde or ketone or compounds which produce them on hydrolysis.

Example: Glucose, fructose, starch, cellulose, glycogen etc.

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